

UPSTAGE DOCUMENT AGENT

From notice analysis to eligibility — Moabora

Drop any announcement. The agent reads the brief, cross-checks your profile, and answers: **can you apply?**

JunctionX Korea 2026

Upstage Track

Team Aquarium

01 UNDERSTAND

PDF · HWP · Poster Ingest

02 JUDGE

Automated Eligibility Verdict

03 ACT

Solar Fit Score & Checklist

Notices are everywhere. Eligibility is not.

Students miss vital opportunities not from lack of listings, but from the exhausting cost of manual reading.

USER SURVEY · N=20 STUDENTS (AGE 20–21)

"What is most painful when checking notices?"

Multiple-choice survey on real notice search bottlenecks.

70% 14 / 20 students
Opening attachments & scanning long briefs (HWP/PDF)

65% 13 / 20 students
Verifying buried qualification clauses in dense text

55% 11 / 20 students
Finding opportunities matching personal profile

Empirical Validation: Format, Eligibility & Matching Friction

PAIN 01 · 70% OF STUDENTS

Unstructured PDF, HWP, & Poster Formats

Scattered across attachments and image posters that search engines and bulletin boards cannot parse.

PAIN 02 · 65% OF STUDENTS

Buried Qualification Clauses in Multi-Page Briefs

Crucial rules ("under 29", "no 4th years", "IT only") hidden in dense guidelines, discovered only after full reading.

PAIN 03 · 55% OF STUDENTS

15–30 Min Matching Overhead per Opportunity

Manual cross-checking across dozens of opportunities wastes hours of repetitive work on ineligible competitions.

The bottleneck is reading, not listing

Existing aggregators stop at collecting links. The real friction begins after you click.

EXISTING NOTICE BOARDS

List Aggregators (Wevity, ContestKorea)

DOCUMENT ACCESS

Hands you raw attachments. You must download and read HWP/PDFs yourself.

ELIGIBILITY CHECK

100% manual check. User spends 15–30 minutes deciphering complex clauses.

RECOMMENDATION

Generic category tags with zero awareness of your actual skills or profile.

TIME TO FIND 5 MATCHING NOTICES (5 BETA TESTERS)

~23 Minutes · Manual skimming across 15+ tabs and attachments

Outcome: 23 min wasted on disqualified briefs

MOABORA DOCUMENT AGENT

Autonomous Document Agent

DOCUMENT ACCESS

Upstage AI reads & structures HWP, PDF, and poster images automatically.

ELIGIBILITY CHECK

Instant 6-rule matching (**Eligible** **Review** **Ineligible**) with explicit reasons.

RECOMMENDATION

Solar Pro 4 calculates personalized fit score (0–100) + tailored rationale.

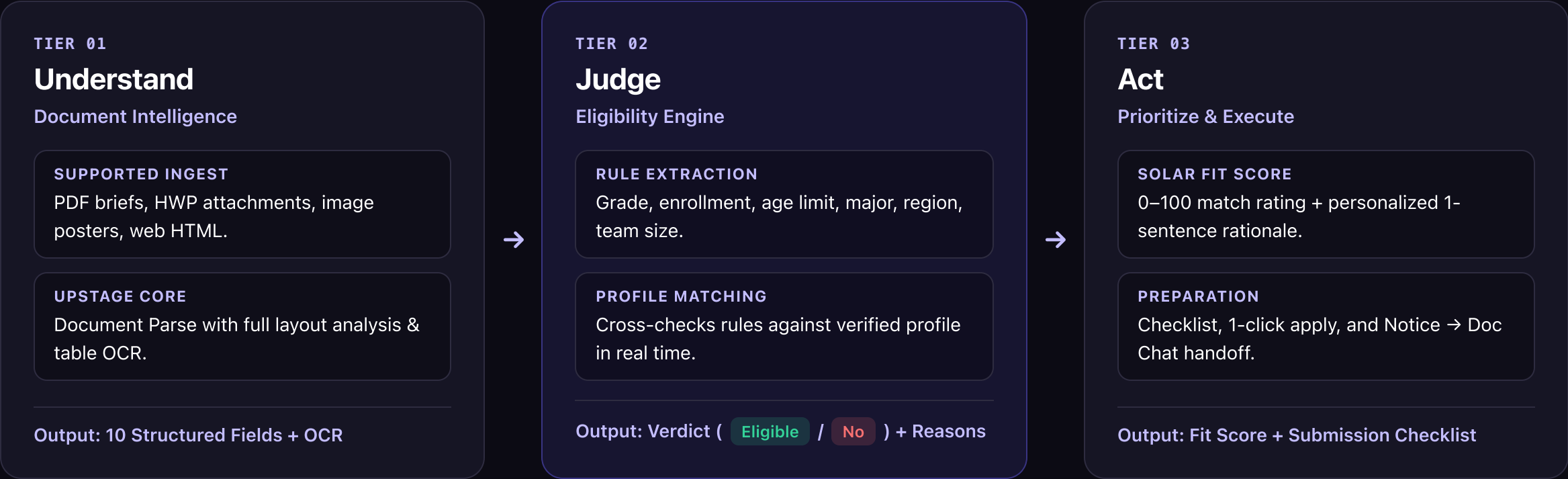
TIME TO FIND 5 MATCHING NOTICES (5 BETA TESTERS)

~5 Minutes · 78% time saved (4.6× faster) via auto-verdicts & Solar fit

Outcome: Instant Verdict & Submission Checklist

Understand → Judge → Act

A three-tier pipeline transforming raw announcement files into verified application actions.



From zero to application in 4 steps

A frictionless 2-minute workflow from profile setup to finalized application decision.

STEP 01 · 1-MINUTE ONBOARDING

Profile Creation

- **Multi-Channel Input:** Certificate (UIE), GitHub/blog link, manual fields, or short note (Solar).
- **Unified Storage:** Merges university, year, status, skills, and interests into a single profile.

Deliverable: Verified Student Profile

STEP 02 · MULTI-CHANNEL INGEST

Notice Collection

- **Flexible Ingest:** Drag & drop PDF / HWP files, paste announcement URLs, or 1-click crawler.
- **Automated Crawling:** Direct collection from Wevity and ContestKorea with HWP extraction.

Deliverable: Multi-Modal Ingestion Pipeline

STEP 03 · PERSONALIZED RANKING

Feed & Fit Scoring

- **Instant Verdicts:** Live eligibility badges (Eligible Review Ineligible) on every card.
- **Solar Pro 4 Fit:** Tailored fit score (0–100) with personalized match rationale.

Deliverable: Intelligent Ranked Opportunity Feed

STEP 04 · PREPARATION & ACTION

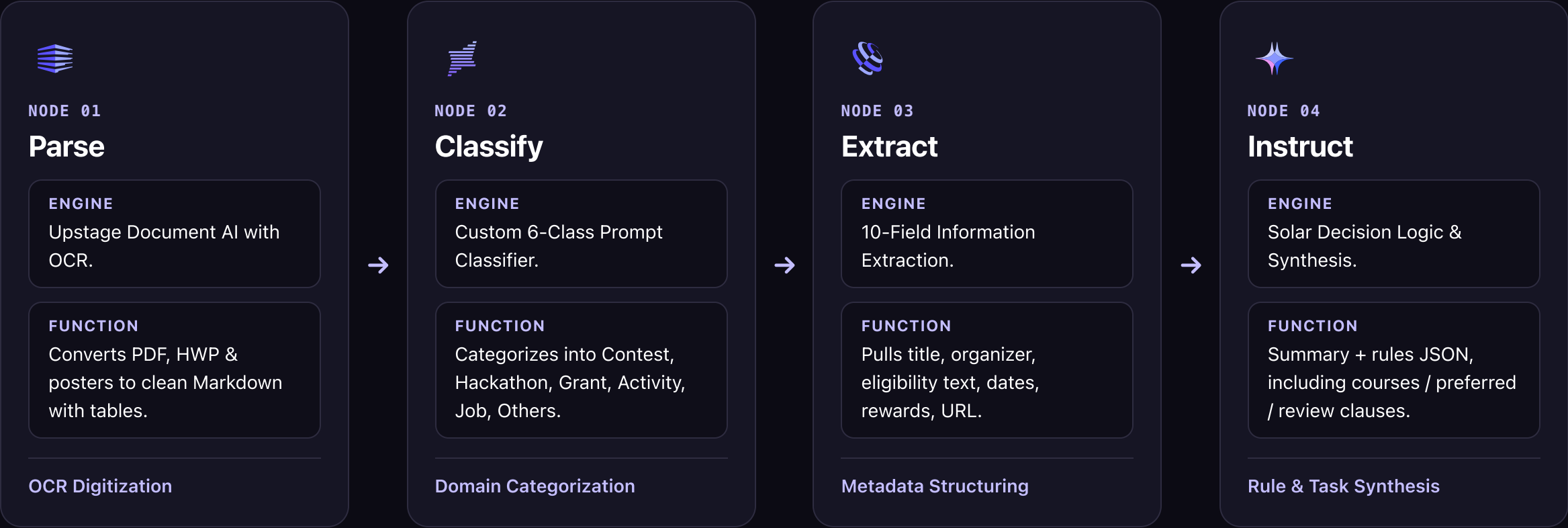
Execution & Doc Chat

- **Review Checklist:** Ambiguous clauses flagged for human verification before apply.
- **Context Handoff:** 1-click transition to Document Chat with source file attached and prompts filled.

Deliverable: Verified Application Submission

Every notice flows through one 4-node agent

All ingest paths (upload, link, crawler) execute the Studio Agent via REST API with zero bypass.



All four product lines in production

Comprehensive integration across Upstage's Document AI and LLM ecosystem.



Studio Agents

PRIMARY ROLE
Core 4-Node Notice Processing Pipeline (`agt\_...`).

ARCHITECTURE
Accumulating snapshot map ensures full node visibility.

REST /v2/responses



Document Parse

PRIMARY ROLE
High-Precision OCR & Layout Analysis.

INTEGRATION
Powers `parse\_document` tool in Document Chat.

/v1/document-digitization



Information Extract

PRIMARY ROLE
Zero-Shot Schema-Driven Document Extraction.

INTEGRATION
Certificate parsing & `extract\_information` chat tool.

/v1/information-extraction



Solar Pro 4

PRIMARY ROLE
Decision Intelligence & Multi-Tool Orchestration.

INTEGRATION
Fit scores, link/note profile extract, streamed tool-calling.

Direct API (OpenAI-Compatible)

Service-grade reliability & transparency

Engineered to handle real-world document edge cases and provide full visibility into AI execution.

ROBUSTNESS

JSON Sanitization

Pre-processes double-encoded Studio Instruct outputs and strips citation markers ([+1]) seamlessly.

Parser Resilience

STATE MANAGEMENT

Snapshot Accumulation

Maintains an accumulating map across Studio polling snapshots so intermediate node outputs are preserved.

Full Node Visibility

TRANSPARENCY

Live NDJSON Streams

Streams pipeline stages, intermediate artifacts, and Solar reasoning tokens in real time to the client.

Streamed Reasoning

RELIABILITY

Resumable Sessions

Server-side workflow sessions survive browser refreshes, preventing duplicate execution and billing.

Refresh-Safe Architecture

OPTIMIZATION

Dual-Layer Caching

SHA-256 document parse cache + profile/notice hash-keyed Solar recommendation cache avoids redundant calls.

Cost & Latency Control

ACCURACY

Deterministic Matcher

Evaluates 6 criteria (grade, status, age, major, region, team) with clear per-rule reason breakdown.

Explainable AI Verdict

3-day hackathon execution & future milestones

Prioritized a rock-solid end-to-end loop while laying an extensible foundation for production scale.

SHIPPED MVP

Delivered in 3 Days

- **3 Ingest paths:** Files, URLs, 2-site Crawlers.
- **4-node Studio Agent:** Full error handling.
- **3 profile paths:** Link, certificate, manual + Solar.
- **Eligibility engine:** 6 rules + Review checklist.
- **NDJSON streams & Chat:** Notice handoff + 3 tools.
- **Supabase Cloud:** Full Docker deployment.

100% Production Ready

CONSCIOUSLY SCOPED OUT

Deferred Complexity

- **Multi-Tenant Auth:** Single-profile demo model.
- **Push Notifications:** Excluded email/push daemons.
- **Scheduled Crawlers:** Manual trigger crawling only.

Avoided Low-Demo Bloat

NEXT MILESTONES

Production Roadmap

- **Multi-User Auth:** Supabase Auth + RLS ready.
- **Proactive Alerts:** D-3 deadline telegram alerts.
- **Application Drafts:** Solar statement writer.
- **Team Matching:** Connect fellow applicants.

Clear Scaling Path

Four bets we actually shipped

Depth of the stack, a complete loop, a sharper problem, and honest scoping — not a demo shell.

Upstage Technology

- Custom 4-node Studio Agent (`agt\_...`) via REST v2 APIs without bypass.
- Production deployment of all 4 Upstage product lines.
- Robust handling of double-encoded JSON and live streaming reasoning.

Max Technical Complexity

Service Completeness

- End-to-end loop: Landing → Profile → Crawl/Ingest → Feed → Detail → Chat.
- Live data from Wevity & ContestKorea with real HWP attachments.
- Resumable workflow sessions that survive browser refresh.

Full Production Loop

Idea Creativity

- Shifting paradigm from "Listing links" to "Delivering application verdicts".
- Reads HWP attachments and posters that traditional aggregators skip.
- Transparent UX displaying live agent thought process and rule proofs.

Novel Problem Framing

Product Planning

- Clear problem focus: Relieving student document fatigue.
- Thoughtful MVP scoping with explicit trade-offs and roadmap.
- Fully documented architecture, data models, and RLS-ready schema.

Execution Rigor

LIVE DEMO

Try Moabora live in production

Fully deployed autonomous document agent running on Supabase Cloud & Upstage Studio Agent APIs.

• LIVE PRODUCTION SERVICE

[https:// moabora.sharp0802.com](https://moabora.sharp0802.com)

Experience the full document intelligence workflow live on web

The agent reads the brief. You just apply.

 Powered by Upstage